

AIPP PROFESSIONAL PROGRAM

Capstone Project Evaluation Framework

ACADEMIC ASSESSMENT ROADMAP

@iLabAfrica | Strathmore University

Academic Year 2026 | Session: June

*Referencing standard academic protocols detailed in: "**Capstone Project Evaluation Framework.pdf**"*

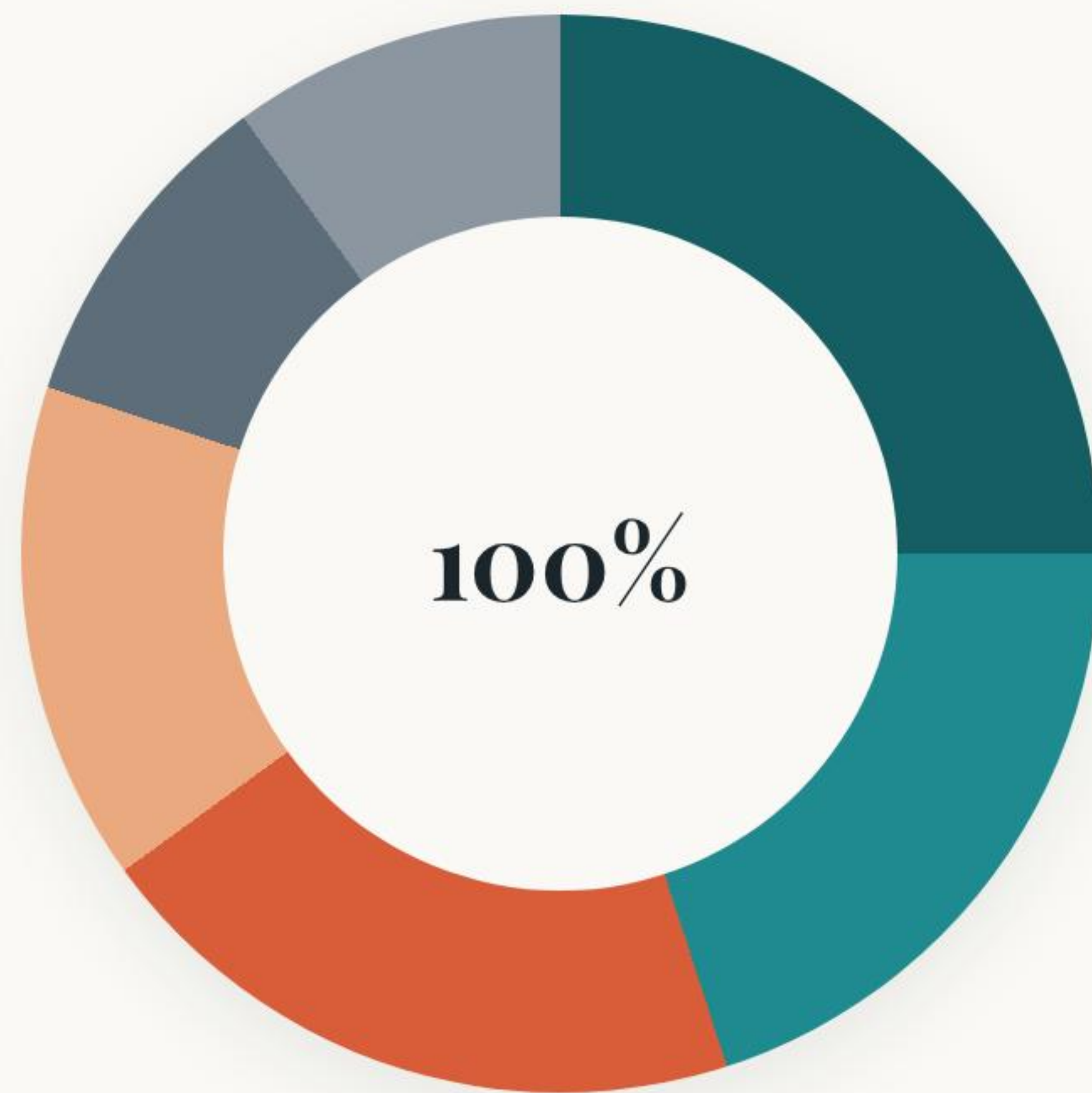
| Program Objectives

The Capstone Vision

The ultimate demonstration of your journey from theory to responsible AI practice, translating deep conceptual modules into functional, secure, and production-ready architectures.

-  **Real-World Value:** Solve complex organizational challenges.
-  **Mastery:** Demonstrate functional Python & Agent engineering.
-  **AI Ethics:** Implement robust risk framework mitigations.
-  **Tangible Impact:** Quantifiable business and technical outcomes.

| Assessment Weights Breakdown



- Technical Execution (25%):** Code execution, dashboard and deployments.
- Problem Framing (20%):** Analytical scoping & business justification.
- AI Ethics & Governance (20%):** Bias mitigation & systematic compliance.
- Integration of Learning (15%):** Modules bridging theory to artifact.
- Communication (10%):** Technical report depth & visual deck logic.
- Reflection (10%):** Honest analysis of architectural limits.

| Problem Framing

Defining the "Why"

Evaluation focuses heavily on mapping out structured, real-world industry pain points where an AI system is deeply justified.

- 🎯 **Impact:** Addressing a defined organizational friction point.
- 🛡️ **Justification:** Defending why AI agents are the correct solution.
- 📅 **Feasibility:** Scoped accurately for a 12-week design cycle.

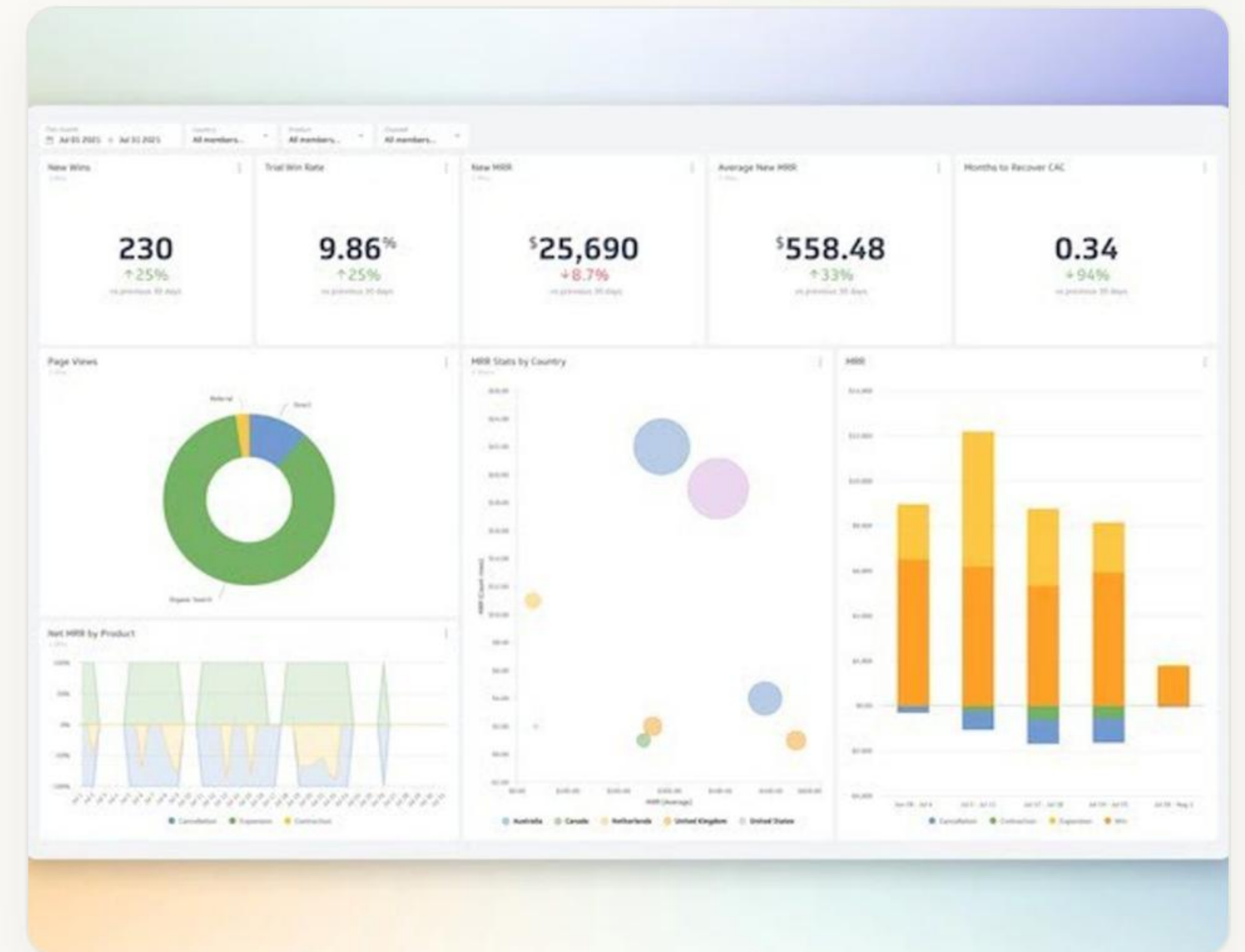


| Technical Execution

Functional Artifacts (25%)

Your technical execution assessment hinges directly on code sophistication, architectural design, and modern deployability principles.

- 🔌 **Clean Python Core:** Modular logic and robust dependency management.
- 🔧 **Prompt Engineering:** Context optimization and robust guardrails.
- 📦 **Deployment Standard:** Functional API or public live UI mockup.
- 📊 **Dashboard UI:** Practical dashboard displaying core metrics.



| Ethics & Governance



Risk ID

Rigorous cataloging of systemic risks including data privacy limits, bias vectors, and performance decay.



Mitigation

Applied technical constraints, custom system guardrails, and validation strategies to preserve data safety.



Explainability

Ensuring system behaviors, algorithmic weights, and output logs are auditable, clear, and transparent.

| Learning Integration

Holistic Application

Capstone projects must showcase meaningful application of theoretical framework structures from the syllabus into concrete programming features.

"Explicit ties to course principles must directly augment the final practical outcomes."

Technical Module Targets

Students are expected to weave logic maps across three foundational program areas:

- 🔗 Deep system models & intelligence.
- Functional backend computation engines.
- 🤖 Multi-agent workflow automation models.

| The Curriculum Blueprint



Python Programming

Creating core analytical loops, handling modern APIs, structured logging, and orchestrating robust computational workflows.



AI Agents

Deploying advanced memory architectures, orchestrating autonomous agentic pipelines, and tracking tooling output.



AI Foundations

Deep algorithmic grounding in neural configurations, transformer frameworks, and mathematical training logics.

Cross-Disciplinary Requirement: Prompt Design frameworks plus Ethical Governance must be visible in the code design.

| Communication & Presentation



Professional Presence (10%)

A successful defense demands matching a polished digital presentation with clear, executive-level technical articulating skills.



Academic Report: Comprehensive written document.



Visual Deck: High contrast, cleanly logic-driven presentation slides.



Technical Q&A: Agile defense of architectural choices.

| Reflection Core Pillar

"A specific, highly honest architectural assessment of what design strategies succeeded, what structural limits failed, and clear operational milestones outlining the next steps of the project."

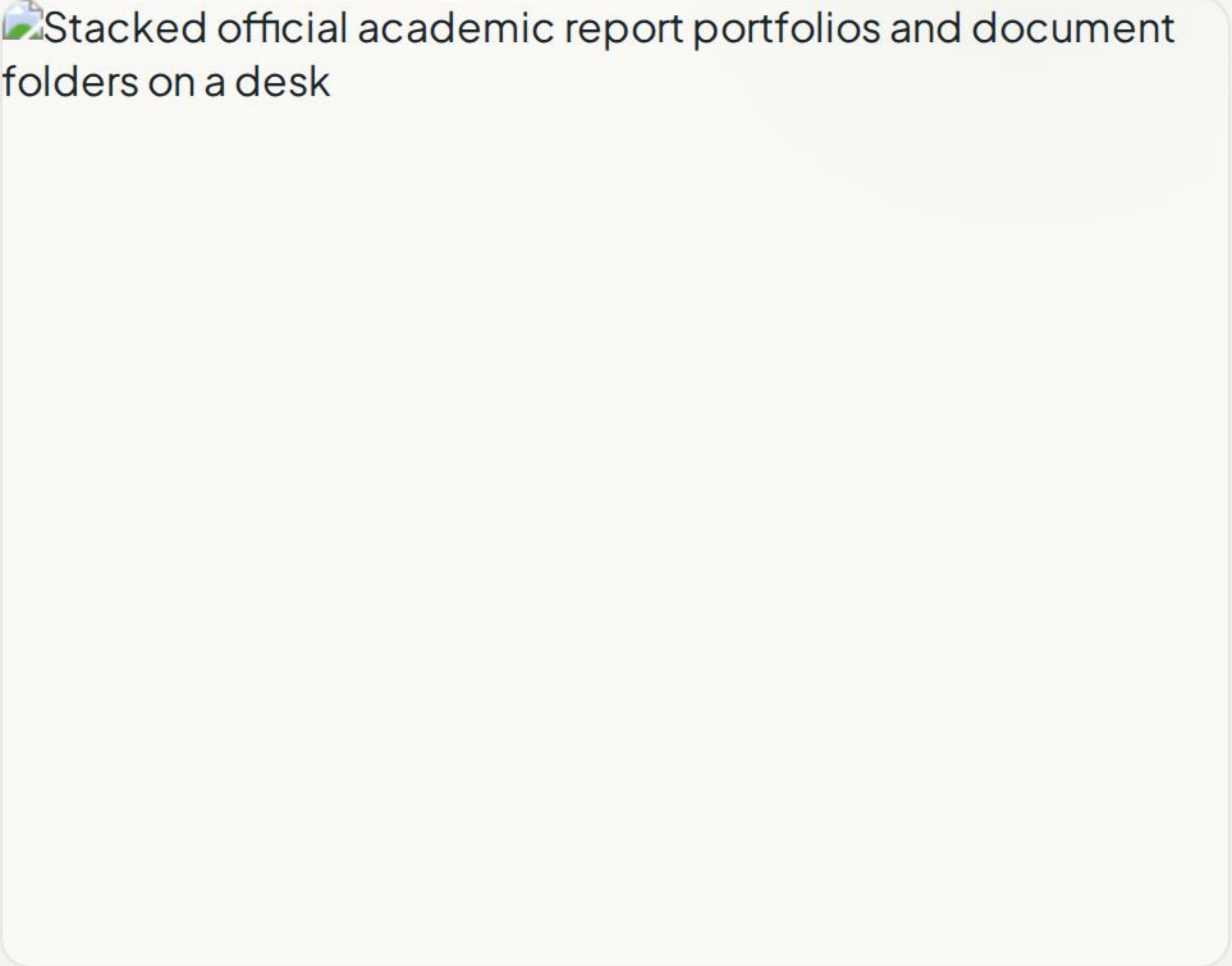
— 10% ACADEMIC REFLECTION STANDARD

| Required Deliverables

Evaluation Portfolio Checklist

Final grading is predicated on submitting a structured, end-to-end technical package.

- ✓ **The Technical Artifact:** Functioning Python code & Prompt logs.
- ✓ **The Document:** Deep executive portfolio report (PDF).
- ✓ **The Presentation:** Modular, highly visual HTML/PDF slides.
- ✓ **The Target System:** Live working prototype or clear screencast.
- ✓ **The Written Reflection:** Self-reflective 500–word essay.



Stacked official academic report portfolios and document folders on a desk

| Grading Scale & Outcomes

SCORE RANGE	ACADEMIC GRADE	REQUIRED ACADEMIC STANDARD
90% – 100%	PASS (MERIT)	Exceptional design rigor visible in both functional code and compliance frameworks.
70% – 89%	PASS (MERIT)	Strong project highlighting consistent, structured real-world deployment goals.
50% – 69%	PASS	Satisfactory evidence of programming application and functional model design.
0% – 49%	FAIL	Missing core deliverables, critical functional gaps, or minimal effort shown.

SYSTEM EXPECTATIONS

Defining Academic Success

"Success is measured not by simple accuracy outputs, but by the thoughtful, ethical integration of theory into real-world, sustainable practice."

| Project Evaluation Roadmap



Week 8

Framing & Scope

Organizational problem selection
approval



Week 10

Prototype Review

Functional Python check & initial risk
matrices



Week 12

Submission

Complete Python code and final technical
report due



Final Period

Defense Panel

Oral presentations and rigorous technical
review

Questions & Discussion

AI PP Professional Program Capstone Framework



Strathmore University



@iLabAfrica Research Center

Source reference: Verbatim criteria defined in "Capstone Project Evaluation Framework.pdf"

| Image Sources



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